

Dataset Card: Bobcat, Coyote, and Gray Fox Scat Traits

Sam Freund

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Summary Information

Field	Description
Name	Dataset_BobcatsCoyotesFoxes-1.csv
Curation Date	Original field collection: 2011–2013; card prepared: March 2026
Sensitivity Level	Low (wildlife observations, no personal data)
Summary	110 DNA-verified scat observations from bobcat, coyote, and gray fox with morphological, biogeochemical, and contextual variables.
Source	Reid (2015), Wildlife Biology, DOI: 10.2981/wlb.00105

Card Authors

Name	Email
Sam Freund	freunds@msoe.edu

Data Overview

Field	Value
Storage Size	9,366 bytes (9.4 KB)
Number of Rows	110
Number of Features	19
Data Format	Comma-separated values (CSV)
Target Label	Species (Bobcat, Coyote, GrayFox)

Numerical Features Summary

Feature	Count	Mean	Std	Min	25%	Median	75%	Max
Length	110	9.30	3.44	2.50	6.50	9.00	11.50	20.50
Diameter	104	18.56	3.88	7.80	16.08	18.05	21.33	30.00
Taper	93	27.43	15.06	2.30	17.30	25.80	37.40	91.50
TI	93	1.60	1.01	0.23	0.99	1.43	1.89	8.68
Mass	109	12.46	8.85	0.94	5.66	9.75	17.61	53.70
d13C	108	-26.86	2.18	-29.85	-28.08	-27.47	-26.45	-19.67
d15N	108	7.44	3.02	1.84	5.62	6.89	8.30	18.00
CN	108	8.40	3.66	4.50	6.20	7.25	8.65	23.60
Year	110	2011.94	0.71	2011	2011	2012	2012	2013

Feature	Count	Mean	Std	Min	25%	Median	75%	Max
Age	110	3.35	1.37	1	3	3	5	5
Number	110	2.62	1.43	1	2	2	3	7
Ropey	110	0.56	0.50	0	0	1	1	1
Segmented	110	0.56	0.50	0	0	1	1	1
Flat	110	0.05	0.23	0	0	0	0	1
Scrape	110	0.05	0.21	0	0	0	0	1

Categorical Features Summary

Feature	Unique Values	Most Common Value	Count
Species	3	Bobcat	57
Month	9	November	17
Site	2	ANNU	92
Location	3	Middle	47

Field Information

Feature	Data Type	Category	Description
Species	categorical	Not a trait (label)	DNA-verified species class.
Month	categorical	Contextual	Month of collection.
Year	integer	Contextual	Year of collection (2011–2013).
Site	categorical	Contextual	Field site code.
Location	categorical	Contextual	Position on trail/road (edge/middle/etc.).
Age	integer	Contextual	Estimated scat age class at collection.
Number	integer	Contextual	Number of pieces in the sample.
Length	float	Morphological	Scat length.
Diameter	float	Morphological	Scat diameter.
Taper	float	Morphological	Length/extent of tapering.
TI	float	Morphological	Taper index.
Mass	float	Morphological	Dry mass of sample.
d13C	float	Biogeochemical	Carbon isotope ratio.
d15N	float	Biogeochemical	Nitrogen isotope ratio.
CN	float	Biogeochemical	Carbon-to-nitrogen ratio.
Ropey	binary (0/1)	Morphological	Rope-like texture indicator.
Segmented	binary (0/1)	Morphological	Segmented morphology indicator.
Flat	binary (0/1)	Morphological	Flat shape indicator.
Scrape	binary (0/1)	Morphological/contextual	Presence of scrape mark near scat. Pres

Example Entries

Representative rows (one per species):

- Coyote sample: January 2012, YOLA edge, length 9.5, diameter 25.7, mass 15.89, d13C -26.85, d15N 6.94, CN 8.5.

- Bobcat sample: January 2012, YOLA middle, length 9.0, diameter 18.8, mass 8.40, d13C -28.73, d15N 8.52, CN 8.1.
- Gray fox sample: January 2012, ANNU middle, length 8.0, mass 2.51, d13C -25.79, d15N 7.83, CN 20.5.

Exploratory Charts

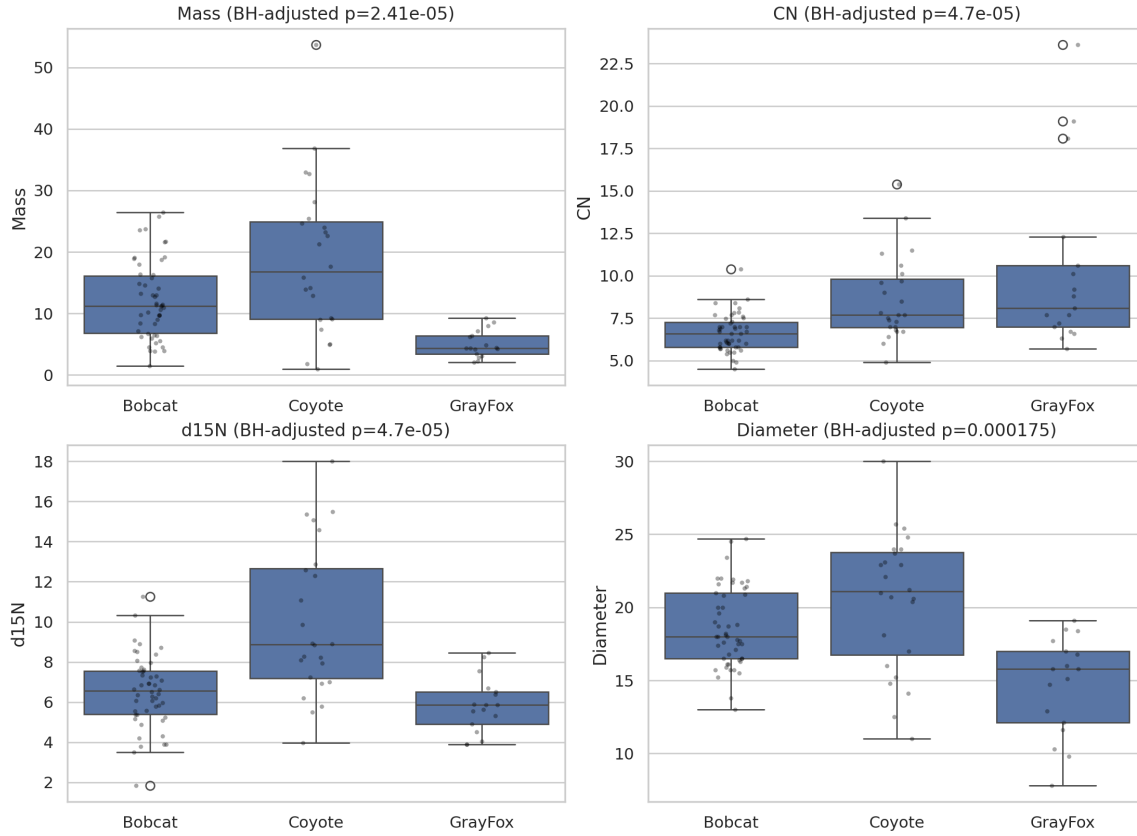


Figure 1: Boxplots of the strongest continuous trait differences by species (based on Kruskal-Wallis tests with BH correction).

Notable Feature Processing

- Data types were explicitly assigned: categorical (Species/Month/Site/Location), integer (Year/Age/Number/binary traits), and float (continuous trait variables).
- Missing values were present in Diameter (6), Taper (17), TI (17), Mass (1), d13C (2), d15N (2), and CN (2).
- For inferential analysis, complete-case rows for tested traits were used (listwise deletion on required columns).
- Outliers were retained because extreme values may reflect real dietary/ecological variability and robust nonparametric tests (Kruskal-Wallis, Mann-Whitney) were used.

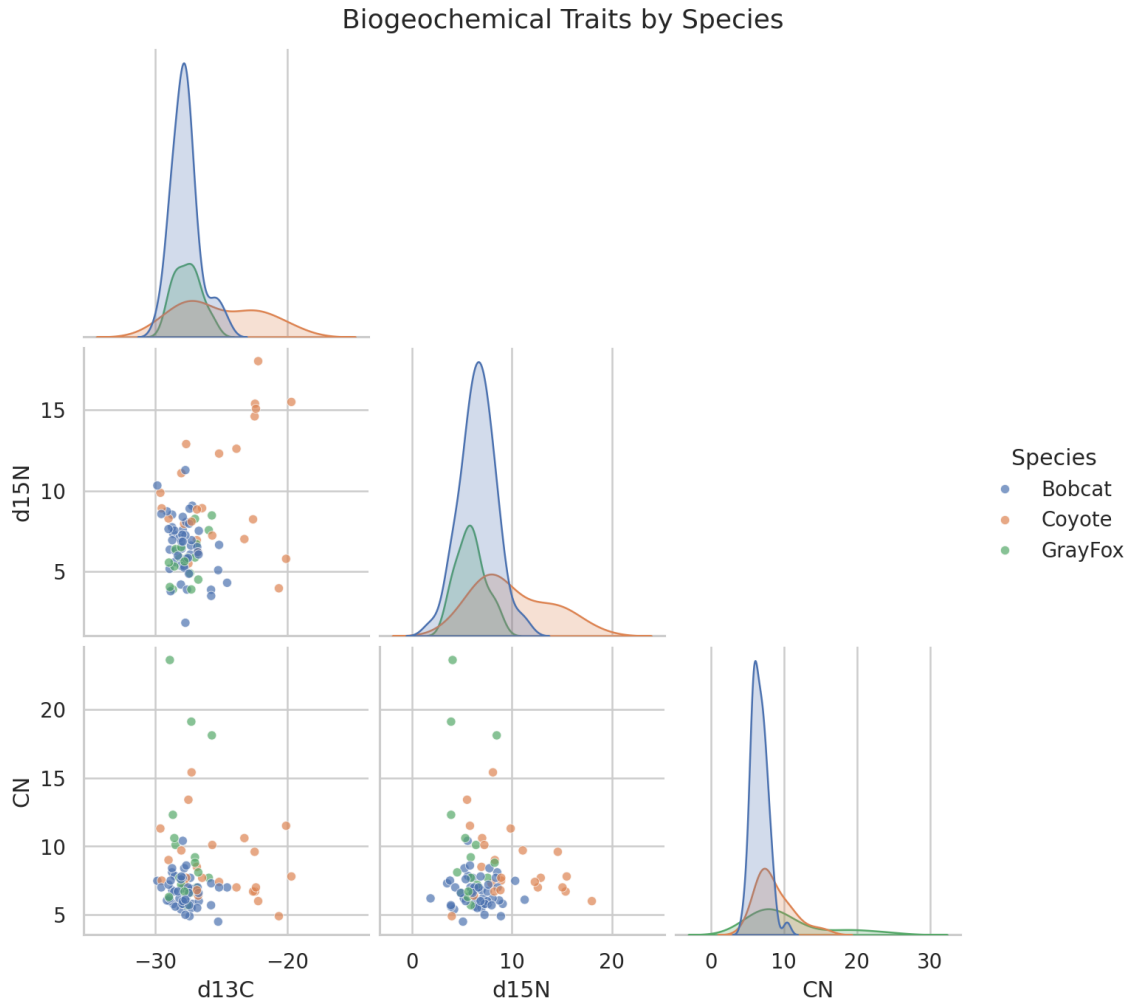


Figure 2: Pairwise relationships among biogeochemical variables (d13C, d15N, and C:N) by species.

Notes

- Species classes are imbalanced (Bobcat $n=57$, Coyote $n=28$, GrayFox $n=25$), so medians and robust tests are emphasized over means alone.
- Contextual variables (site/month/location/year) are useful for collection metadata but were excluded from the species-discrimination hypothesis tests.
- The dataset supports strong comparisons for several traits (mass, diameter, d13C, d15N, C:N, segmented), but results are site- and time-specific to central coastal California.

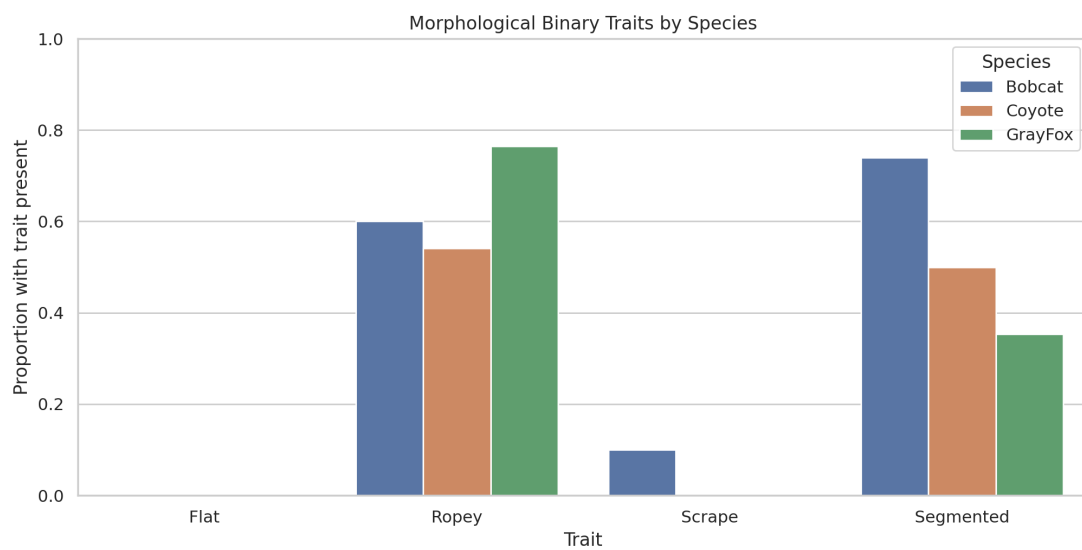


Figure 3: Prevalence of binary morphological traits by species.